

FLOODGUARD STO. NINO RIVER

Complete Defense Record and Evidence Guide

Final operational-model rationale | transparent OLS MLR time-series forecasting

Item	Verified position
Final model	Candidate 9 (C9)
Equation inputs	Sto. Nino WL(t-1), Sto. Nino WL(t-3), Mt. Boso-Boso Rain(t-1)
Method	OLS Multiple Linear Regression; exact-calendar lags; past inputs only
Decision status	CERTIFIED_PASS

Plain-language answer: C9 uses the river level yesterday, the river level three days ago, and prior-day Boso-Boso rainfall to forecast the next station water-level value. It is an operationally balanced model, not a claim of universal storm superiority.

How to use this record

- Sections 1-5 explain the candidate-selection process.
- Sections 6-8 explain validation, persistence, event evidence, and runtime safety.
- Section 9 gives ready-to-say panel answers.

1. What the Model Does

Sto. Nino is a station-specific daily forecasting component. It is designed to support one-day-ahead monitoring using its audited historical water-level and rainfall inputs.

Question	Answer
Forecast horizon	One calendar day ahead, using only prior-day information
Method boundary	Transparent OLS MLR time series; no AI, black-box model, Ridge, or hidden regularization
Lag rule	Exact calendar dates, not the preceding spreadsheet row
Missing-data rule	No zero imputation, interpolation, forward-fill, or backward-fill
Safety boundary	Forecast supports monitoring. Live observations and official warning protocols remain the safety authority.

Why it is a time-series MLR model

C9 has three predictors, including two past water-level lags. The lags carry river memory; prior-day Boso-Boso rainfall adds an upstream rainfall signal. $K = 3$, so it is Multiple Linear Regression and a time-series model.

2. The Frozen Selection Process

Candidates were defined before validation performance was used to select the operational model. The selection procedure therefore evaluates generalization on later chronological dates rather than choosing a model because it looked good on a known storm.

Step	Rule
1. Structure	Candidate must use past-day predictors only and meet the project MLR requirement.
2. Reliability	VIF must be ≤ 5.0 and required slopes must pass training-sample HC3 robust inference.
3. Validation	Fit on earlier chronology and compare frozen candidates on later unseen chronology.
4. Fair comparison	Use exact common validation dates for pairwise comparisons.
5. Deployment	Consider sensor/input availability, persistence benchmark, and safe fallback separately.

Panel line: “We selected under a pre-defined overall validation process. We did not reselect the model by looking only at a few named typhoons after the fact.”

3. Why Candidate 9 (C9) Was Chosen

C9 was selected as the operational prototype because it satisfied the statistical and model-form requirements, used a practical input footprint, had stronger availability than the main alternative C15, and showed strong examined critical-event results against C15 and true persistence. Candidate 13 was the development-validation tournament winner; C9 was not falsely relabeled as that winner.

Selection dimension	Evidence
Model form	K = 3 OLS MLR time series: two WL lags plus Boso-Boso rain lag.
HC3 / VIF	C9 passed the package diagnostic screens; all production slopes are retained under the documented governance.
Validation basis	C13 won the development validation tournament. C9 was selected for operational balance, not claimed as universal validation winner.
Operational basis	C9 availability 617/681 (90.60%) versus C15 557/681 (81.79%) in the stated retrospective comparison.

What this decision does not claim

- C9 was not the all-candidate validation winner.
- C9 was not proven best against every candidate on every extreme event.
- C9 does not guarantee a typhoon peak or replace official warnings.

4. HC3 and VIF in Simple Terms

OLS creates the equation. HC3 is a cautious uncertainty check: it asks whether a predictor still appears dependable when ordinary days and unusual large-error days do not behave equally. VIF checks whether two inputs are so similar that the separate coefficients become unstable.

Check	Question it answers	Effect
HC3	"Can we still trust this predictor after a cautious error calculation?"	Does not change the fitted OLS prediction equation; it changes the uncertainty and eligibility evidence.
VIF	"Are two predictors too redundant?"	Prevents unstable coefficients caused by overlapping inputs.

HC3 is a quality gate, not the objective function. It does not make a forecast weaker or stronger; the equation is still ordinary least squares.

5. Chronological Validation and Persistence

The tournament and persistence records are kept separate. The following retrospective availability evidence explains why C9 was preferred operationally over the less available C15 alternative.

Comparison	C9	C15	True persistence
Availability on stated retrospective basis	617 / 681 = 90.60%	557 / 681 = 81.79%	650 / 681 = 95.45%
Role	Operational prototype	Practical comparator	Deterministic benchmark / fallback

Why persistence is shown separately

True persistence is a deterministic benchmark: $\text{prediction}(t) = \text{observed water level}(t-1)$. It is not the same as a fitted AR1 regression, which has an estimated intercept and slope.

Persistence achieved the best stated overall retrospective MAE of 0.164255 m on 557 exact-common days. This is disclosed; it is not hidden. C9 is justified by the full operational balance and the examined event evidence, not by claiming persistence loses every metric.

6. Extreme-Day and Event Evidence

The existing event comparison is C9 versus C15 versus true persistence. It is meaningful operational evidence, but it is not an all-candidate event tournament.

Examined subset	C9 MAE	C15 MAE	Persistence MAE	Lowest MAE
Named typhoon window	0.7266	0.7654	0.8367	C9
Alert subset	0.8746	1.0360	1.2800	C9
Rapid-change subset	1.0339	1.1911	1.4496	C9
Rapid-rise subset	1.2806	1.3400	1.4411	C9
Operational-critical union	0.9755	1.1168	1.3610	C9

Correct interpretation

C9 had the lowest MAE in the examined common event subsets against C15 and persistence. It also has a recorded major failure: on 21 Jul 2025, following a +4.80 m rise, C9 missed by 4.8053 m, essentially matching persistence error. This limitation must be stated plainly.

Do not say the selected model is proven best for every typhoon. The candidate is the most defensible overall operational choice under the frozen methodology; event evidence is reported as a separate safety check.

7. Deployment and Fallback

Runtime condition	Output
Required C9 inputs are valid	Calculate Candidate 9 OLS MLR forecast.
Required rainfall input missing; prior Sto. Nino level available	Use the documented station fallback/monitoring rule; do not invent rainfall.
Required prior Sto. Nino water level missing	Return forecast unavailable; no imputation.

Why this is safe

- The model never invents a missing prior water-level value.
- Persistence is an explicit deterministic fallback, not a hidden substitute model.
- The system must show the forecast source/fallback state so users know what was used.
- A forecast supplements, never replaces, live monitoring and official warnings.

8. What is Recorded as Proof

Evidence item	What it proves
Frozen candidate definitions and checksum	Candidates were specified before selection.
Model-ready data and exact-calendar-lag audit	No row-shift lag leakage; missing values retained.
OLS, HC3, and VIF tables	Equation, uncertainty, and collinearity evidence.
Validation predictions and pairwise comparisons	Later-date performance and fair common-date comparison.
Persistence and event-subset files	Benchmark trade-offs and stated extreme-day evidence.
Formula-backed Excel workbook	Visible native calculation trail with formula-error scan.
Runtime engines, parity tests, manifest, checksum	Deployment equation matches the audited model and is reproducible.

The package is the source of truth. This defense record summarizes that evidence and never replaces the CSV, workbook, or reproducibility trail.

9. Panel Q&A;: Say It Simply

Likely question	Answer
Why this model?	C9 is the operationally balanced choice: valid MLR form, practical availability, and stronger examined event results versus C15 and persistence.
Did HC3 pick it by itself?	No. HC3 and VIF were gates. The final choice then followed frozen chronological validation and operational rules.
Is it best in every typhoon?	No. We do not make that claim. Extreme-day results are separate, subset-specific evidence.
Why not select using only typhoon days?	That would be hindsight selection on a small, known set of events and could overfit those events.
What happens when an input is missing?	No input is fabricated. If required inputs are unavailable, the runtime follows its explicit fallback/unavailable rule.
What protects the public?	Live measurements, official warning protocols, and human/agency decision-making remain the safety authority.

Closing defense position: “Our model is not presented as perfect for every storm. It is the most defensible station-specific OLS MLR operational model under a frozen, transparent, reproducible selection process.”